

M13 Batch 10 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 918–1008

GPWS modes and windshear, enhanced GPWS and terrain awareness, vibration monitoring and the fuel quantity indicating system.

227 questions · Level 3: 80 · Level 2: 104 · Level 1: 43

● marks a question built on a fact your instructor highlighted or underlined in the manual (169 of 227). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

GPWS mode 6, mode 7 and windshear

PDF pages 919–927 · 37 questions

LEVEL 1 — RECALL

1. What makes the EGPWS give Mode 6 alerts and call-outs ●
 - a. The aircraft climbing above predefined altitudes
 - b. The aircraft descending below predefined altitudes
 - c. The aircraft exceeding a predefined airspeed
2. Which indications Mode 6 alerts and call-outs give to the crew ●
 - a. Aural messages and ARINC 429 outputs only
 - b. Visual warnings and aural messages only
 - c. Visual warnings and ARINC 429 outputs only
3. Which flight phases the Mode 7 windshear alerting covers ●
 - a. Cruise and initial descent
 - b. Take-off and final approach
 - c. Taxi and initial climb
4. Between which heights above ground the windshear detection is active ●
 - a. 50 and 2500 feet AGL
 - b. 10 and 1500 feet AGL
 - c. 200 and 1000 feet AGL
5. An airport needs envelope modulation, so how is it held in the GPWC memory? ●
 - a. By their runway heading alone
 - b. By their ICAO four-letter code
 - c. By their latitude and longitude
6. What the terrain awareness alerting function uses to predict a conflict ahead ●
 - a. The radio altimeter signal with the nearest runway length
 - b. The aircraft position and altitude with an internal database
 - c. The weather radar returns with the aircraft ground speed
7. Terrain awareness is working normally, so which two systems supply its aeroplane data? ●
 - a. The ADC and the ILS
 - b. The ADIRS and the GPS
 - c. The FMS and the WXR

8. Which alerting envelopes the terrain alerting function computes ●
- a. A look-up envelope and a lateral offset envelope
 - b. A caution-level envelope and a warning-level envelope
 - c. A take-off envelope and a final approach envelope

LEVEL 2 — UNDERSTAND AND EXPLAIN

9. From where the decision height used by the minimums call-out is taken
- a. A fixed preset inside the radio altimeter
 - b. The setting on the EFIS control panels
 - c. The runway elevation in the terrain database
10. An approaching minimums call-out is selected, so how far above the decision height does it normally come on? ●
- a. 100 feet
 - b. 130 feet
 - c. 80 feet
11. The approaching option is set so that the aural comes at decision height + 100 feet, so which call-out is then given? ●
- a. approaching decision height
 - b. approaching minimums
 - c. plus hundred
12. The windshear pre-alert comes up on take-off, so which indications does it give? ●
- a. Aural outputs and a cockpit siren
 - b. Visual and ARINC 429 outputs
 - c. Aural, visual and ARINC 429 outputs
13. What the measured windshear value represents
- a. The vector sum of inertial and air mass accelerations
 - b. The rate of change of barometric altitude
 - c. The difference between ground speed and airspeed
14. Which further aeroplane parameters the GPWC checks near such an airport ●
- a. Ground speed, roll attitude and calculated flight path angle
 - b. Radio altitude, magnetic track and glide slope deviation
 - c. Vertical speed, true heading and terrain elevation data
15. Which navigational signals the envelope modulation feature can use
- a. Updated VOR and DME range and bearing signals
 - b. Localizer and glide slope beam deviation signals
 - c. GPS or updated flight management system signals
16. Which sources terrain awareness uses for latitude and longitude ●
- a. GPS or WXR
 - b. GPS or FMS
 - c. ADIRS or ADC
17. Which of the terrain awareness input quantities is worked out by the EGPWC itself ●
- a. Barometric altitude
 - b. Ground track
 - c. Flight path angle

18. What the EGPWC compares with its terrain database to find a threat ●
 - a. The aeroplane pitch and roll
 - b. The aeroplane position and track
 - c. The aeroplane speed and flap setting
19. How the terrain display shows terrain height relative to the aeroplane ●
 - a. By solid contour lines
 - b. By numeric height labels
 - c. By different coloured dots
20. The caution level aural has just been heard and the envelope is still penetrated, so when does it come again? ●
 - a. Every seven seconds
 - b. Every three seconds
 - c. Without any pause
21. The aeroplane stays inside the terrain warning envelope, so how is PULL UP given? ●
 - a. Continuously
 - b. Twice and then once per 20% height loss
 - c. Every seven seconds
22. What the alert discretises of the look-ahead function are provided for ●
 - a. To inhibit the weather radar transmitter
 - b. To reset the radio altimeter output
 - c. To energise the terrain display relays
23. By which three dimensions the terrain alerting envelopes are sized
 - a. A look-ahead distance, an altitude offset and a lateral distance
 - b. A ground speed, a flight path angle and a roll attitude
 - c. A runway distance, a radio altitude and a ground track angle

LEVEL 3 — APPLY AND REASON

24. At the height where the call-outs begin the operator rejects the aural TWENTY FIVE HUNDRED, so what is the other option there? ●
 - a. The aural TOO LOW TERRAIN
 - b. The aural MINIMUMS MINIMUMS
 - c. The aural RADIO ALTITUDE
25. The aeroplane is at 300 feet radio altitude and the roll is increased steadily, so at which angles does the call-out occur? ●
 - a. 15, 30 and 45 degrees
 - b. 35, 40 and 45 degrees
 - c. 10, 20 and 30 degrees
26. On a low go-around at 60 feet radio altitude the pilot rolls to 20 degrees, so what does Mode 6 do? ●
 - a. It gives the aural MINIMUMS, MINIMUMS
 - b. It gives no roll call-out at that height
 - c. It gives the aural BANK ANGLE, BANK ANGLE

27. At 200 feet radio altitude the aeroplane is rolled to 20 degrees but nothing is heard, so why is that? ●
- Above 130 feet at least 45 degrees is needed
 - Above 130 feet at least 35 degrees is needed
 - Above 130 feet roll call-outs are inhibited
28. A windshear warning is triggered on take-off, so how do its indications differ from those of the pre-alert? ●
- An ARINC 429 output is added
 - A visual indication is added
 - An aural indication is added
29. A shear indication comes up on take-off with no sound at all, so which level of Mode 7 has been reached? ●
- The terrain caution
 - The warning
 - The pre-alert
30. On final approach at 900 feet AGL the head wind increases quickly and a severe up draft is met, so what does Mode 7 do? ●
- It stays quiet, because the gear is not yet down
 - It alerts, because the height is inside the active band
 - It alerts only after the aeroplane is below 500 feet
31. Severe shear is met at 1800 feet AGL in the climb but nothing is annunciated, so why is that? ●
- Detection is inhibited during any climb
 - Detection stops above 2500 feet AGL
 - Detection stops above 1500 feet AGL
32. The aeroplane is near a listed airport but the glide slope deviation and runway heading fail to match, so what does the GPWC conclude? ●
- A modulation condition exists anyway
 - No modulation condition exists
 - The terrain database must be in error
33. Why is the top radio altitude limit of the mode 2 alert area lowered at some locations?
- To let the aeroplane fly closer to the terrain without alerts
 - To let the aeroplane approach without lowering the gear
 - To let the aeroplane descend faster without any alerts
34. An approach needs mode 5 annunciations higher than normal and with the gear still up, so which change makes that possible?
- The alert area is lowered and a gear down requirement added
 - The descent rate limit is increased and the alert area raised
 - The alert area is increased and the gear down requirement removed
35. Both GPS and FMS position data are lost in the cruise, so which EGPWS function can no longer work? ●
- The look-ahead comparison with the terrain database
 - The radio altimeter height measurement function
 - The barometric altitude correction in the ADC

36. An aural is heard and exactly seven seconds later the same words come again, so which envelope is still penetrated? ●
- The clearance floor envelope
 - The warning envelope
 - The caution envelope
37. What happens to the look-ahead distance as the aeroplane accelerates and why
- It stays fixed, because altitude alone sets it
 - It decreases, to keep the envelope narrow
 - It increases, to keep the alerting time about equal

Answers — GPWS mode 6, mode 7 and windshear

- b** Mode 6 is triggered by the aircraft descending below predefined altitudes such as DH and minimums. (L1, PDF p. 919)
- a** Mode 6 produces aural messages and ARINC 429 outputs but no visual indication. (L1, PDF p. 919)
- b** Mode 7 alerts during take-off and final approach. (L1, PDF p. 921)
- b** Detection is active between 10 and 1500 feet AGL. (L1, PDF p. 921)
- c** Those airports are held in memory identified by latitude and longitude. (L1, PDF p. 923)
- b** Geographic position and altitude are compared with the internal terrain database. (L1, PDF p. 925)
- b** Aeroplane data comes from the ADIRS and the GPS. (L1, PDF p. 925)
- b** Two envelopes are computed, one for caution level and one for warning level. (L1, PDF p. 927)
- b** The minimums call-out uses the decision height set on the EFIS control panels. (L2, PDF p. 919)
- c** The approaching minimums call-out normally comes on 80 feet above the decision height. (L2, PDF p. 919)
- c** The plus hundred call-out is the one whose altitude selection is decision height + 100 feet. (L2, PDF p. 919)
- b** The caution, or pre-alert, gives visual and ARINC 429 indications only. (L2, PDF p. 921)
- a** The value is the vector sum of inertial versus air mass accelerations along and across the flight path. (L2, PDF p. 921)
- b** Radio altitude, magnetic track and glide slope deviation are all on the printed list. (L2, PDF p. 923)
- c** The feature uses GPS or updated FMS navigational signals. (L2, PDF p. 923)
- b** Latitude and longitude come from GPS or FMS. (L2, PDF p. 925)
- c** Only the flight path angle is noted as calculated by the EGPWC; the others are received. (L2, PDF p. 925)
- b** Position and track are compared with the database. (L2, PDF p. 925)
- c** Terrain height relative to the aeroplane is shown by different coloured dots. (L2, PDF p. 925)
- a** The caution message repeats every seven seconds while the envelope is penetrated. (L2, PDF p. 927)
- a** PULL UP repeats continuously for as long as the warning envelope is penetrated. (L2, PDF p. 927)
- c** The discretes energise the terrain display relays. (L2, PDF p. 927)
- a** The three sizing dimensions are look-ahead distance, altitude offset and lateral distance. (L2, PDF p. 927)
- c** At 2500 feet the alternative to TWENTY FIVE HUNDRED is the aural RADIO ALTITUDE. (L3, PDF p. 919)
- b** Above 130 feet the bank angle call-out comes at 35, 40 and 45 degrees. (L3, PDF p. 919)
- c** 60 feet lies between 30 and 130 feet and 20 degrees is larger than 10 degrees, so the call-out is given. (L3, PDF p. 919)
- b** Above 130 feet the first bank angle call-out is at 35 degrees, so 20 degrees gives nothing. (L3, PDF p. 919)
- c** The warning adds the aural to the visual and ARINC 429 indications the caution already gives. (L3, PDF p. 921)
- c** Only the warning has an aural, so a silent visual indication is the caution or pre-alert. (L3, PDF p. 921)
- b** 900 feet lies inside the printed 10 to 1500 feet AGL detection band, so Mode 7 gives its alert. (L3, PDF p. 921)
- c** The active band ends at 1500 feet AGL, so 1800 feet is outside it. (L3, PDF p. 921)
- b** Position alone is not enough; the extra parameters must also confirm the condition. (L3, PDF p. 923)
- a** Lowering the mode 2 area lets the aeroplane fly closer to the terrain without alerts. (L3, PDF p. 924)
- c** Mode 5 modulation increases the alert area limits and removes the gear down requirement. (L3, PDF p. 924)

- 35. a Without latitude and longitude there is no position to compare with the terrain database. (L3, PDF p. 925)
- 36. c A seven second repeat is the caution level; the warning level repeats continuously. (L3, PDF p. 927)
- 37. c The distance grows with ground speed so that alerting times stay roughly the same. (L3, PDF p. 927)

Enhanced GPWS and terrain awareness

PDF pages 929–939 · 26 questions

LEVEL 1 — RECALL

38. Which colour paints ground that violates the terrain caution envelope ●
- a. 100 per cent yellow
 - b. 100 per cent red
 - c. 50 per cent magenta
39. Which colour paints ground that violates the terrain warning envelope ●
- a. 100 per cent red
 - b. 100 per cent yellow
 - c. 50 per cent green
40. What the TCF alert function builds around the intended runway ●
- a. An increasing terrain clearance envelope
 - b. A circular windshear detection envelope
 - c. A fixed glide slope deviation envelope
41. Which two sources feed the navigational displays through the select relays ●
- a. The EGPWC and the WXR transceiver
 - b. The ADIRS and the flight management system
 - c. The GPWC and the radio altimeter

LEVEL 2 — UNDERSTAND AND EXPLAIN

42. Under which condition terrain is kept off the display
- a. When it lies more than 2000 ft below reference altitude
 - b. When it lies more than 1000 ft above reference altitude
 - c. When it lies more than 2000 ft above the runway
43. On which three quantities the terrain clearance floor alerts are based ●
- a. Aircraft heading, runway length and barometric altitude data
 - b. Aircraft location, nearest runway centre point and radio altitude
 - c. Ground speed, flight path angle and nearest terrain elevation
44. When the terrain clearance floor is active during take-off
- a. When the flaps have been retracted
 - b. When mode 4 protection is not available
 - c. When the landing gear is still down
45. The aeroplane moves further away from the airport, so how does the height of the clearance envelope change? ●
- a. It stays the same at every distance
 - b. It increases as the distance increases
 - c. It decreases as the distance increases
46. Which aeroplane data the EGPWC compares with the clearance envelope data ●
- a. Heading, ground speed and radio altitude
 - b. Latitude, longitude and barometric altitude
 - c. Latitude, longitude and radio altitude

47. On which data bus standard the terrain and weather display data is carried ●
- ARINC 453
 - ARINC 429
 - ARINC 629
48. Which equipment receives the data from the terrain weather select relays ●
- The flight warning computers
 - The air data computers
 - The display electronics units
49. What happens when the TERR switch is selected a second time
- The relay deenergises and the weather radar is connected
 - The relay deenergises and a test pattern is shown
 - The relay stays energised and the terrain stays shown

LEVEL 3 — APPLY AND REASON

50. The aeroplane is in level flight and an area ahead is painted high density red, so where does that ground lie?
- Between 1000 and 2000 ft above it
 - More than 2000 ft below the aeroplane
 - More than 2000 ft above the aeroplane
51. An area along the track is painted solid yellow rather than in a dot pattern, so what has happened? ●
- The warning envelope has been violated
 - The caution envelope has been violated
 - The clearance floor has been penetrated
52. A QFE approach is flown into a field with an extreme surface temperature, so why does the EGPWS need no special procedure?
- Corrected barometric altitude is used instead
 - The radio altimeter replaces the air data input
 - Its computed altitude copes with both conditions
53. Why the terrain awareness function still works properly after a wrong altimeter subscale setting
- Its computed altitude is not susceptible to that error
 - The radio altimeter replaces every altitude input
 - The air data computer recomputes the corrected value
54. The aeroplane flies through a strong inversion layer, so why can the corrected barometric altitude mislead the EGPWS?
- It freezes at the last valid value
 - It can be in error against the true height
 - It is replaced by uncorrected altitude
55. The landing gear is lowered in the descent, so how does the lower edge of the medium density yellow band change?
- From 500 ft below to 250 ft below the reference altitude
 - From 1000 ft below to 500 ft below the reference altitude
 - From 250 ft below to 500 ft below the reference altitude

56. The landing gear is up and ground lies 400 ft below the reference altitude, so in which band is it painted?
- Medium density yellow
 - Light density green
 - Medium density green
57. TERR is selected on an EFIS control panel, so what does the select relay do?
- It deenergises and connects weather data to the display units
 - It energises and connects terrain data to the display units
 - It energises and connects the hazard bus to the display units
58. What the transfer relays do while the weather radar is on and the caution criteria become fulfilled
- They drop the terrain bus and switch to the weather bus
 - They drop the weather bus and switch to the terrain bus
 - They connect both buses to the display units at once
59. A predictive windshear alert arrives while the terrain picture is up, so which picture is then shown?
- The terrain test pattern
 - The terrain one, with the shear data on it
 - The weather one, with the shear data on it
60. The terrain display relays have been energised by the alert discretes, so what do the navigational displays then carry? ●
- The weather radar picture as before
 - The conflicting ground in yellow or red
 - The terrain test pattern in cyan
61. A siren is followed by the same word three times during the climb-out at 300 feet, so which mode has alerted?
- Mode 5
 - Mode 7
 - Mode 2
62. The short level 1 test aural sequence begins with glideslope INOP, so what does that tell the engineer?
- The glide slope lamp itself has failed
 - No valid glide slope input is present
 - The test was started below 2000 feet
63. A level one test is run from the front panel of the computer, so why is the GPW module used instead for an operational test?
- The front panel test only checks the power supply
 - The front panel test gives out no aural messages at all
 - The flight deck annunciations cannot be seen there

Answers — Enhanced GPWS and terrain awareness

38. **a** Caution envelope violations are painted solid yellow. (L1, PDF p. 929)
39. **a** Warning envelope violations are painted solid red. (L1, PDF p. 929)
40. **a** TCF builds a terrain clearance envelope that grows with distance from the runway. (L1, PDF p. 933)
41. **a** The two sources are the EGPWC and the weather radar transceiver. (L1, PDF p. 935)
42. **a** Terrain more than 2000 ft below reference altitude is not painted. (L2, PDF p. 929)
43. **b** Location, nearest runway centre point and radio altitude are the three inputs. (L2, PDF p. 933)
44. **b** In the take-off mode TCF covers the gap where mode 4 protection is missing. (L2, PDF p. 933)

45. **b** The envelope gets higher the further the aeroplane is from the airport. (L2, PDF p. 933)
46. **c** Latitude, longitude and radio altitude are compared with the envelope data. (L2, PDF p. 933)
47. **a** The display data runs on ARINC 453 buses. (L2, PDF p. 935)
48. **c** The relays feed the display electronics units, which drive the NDs. (L2, PDF p. 935)
49. **a** A second selection deenergises the relay and reconnects the weather radar. (L2, PDF p. 935)
50. **c** In level flight the reference altitude is the aircraft altitude, so high density red is more than 2000 ft above it. (L3, PDF p. 929)
51. **b** Solid yellow is reserved for ground that violates the caution envelope. (L3, PDF p. 929)
52. **c** The geometric altitude function covers temperature extremes and QFE operation together. (L3, PDF p. 931)
53. **a** Geometric altitude is immune to an incorrect altimeter setting, so terrain awareness is unaffected. (L3, PDF p. 931)
54. **b** Under non-standard conditions such as inversions the corrected barometric altitude differs from the true altitude. (L3, PDF p. 931)
55. **a** With the gear up the band reaches 500 ft below, with the gear down only 250 ft below. (L3, PDF p. 932)
56. **a** With the gear up the medium density yellow band runs down to 500 ft below, so 400 ft below falls in it. (L3, PDF p. 932)
57. **b** Selecting TERR energises the relay and connects terrain data to the display units. (L3, PDF p. 935)
58. **b** With the criteria fulfilled the relays swap the weather bus out for the terrain bus. (L3, PDF p. 935)
59. **c** A shear alert overrides the terrain picture and reverts to the weather display. (L3, PDF p. 935)
60. **b** Once the relays are energised the conflicting ground is painted solid yellow or solid red. (L3, PDF p. 935)
61. **b** Only the mode 7 warning begins with a siren and repeats its word three times. (L3, PDF p. 938)
62. **b** The INOP annunciation in the sequence means the glide slope input is missing. (L3, PDF p. 939)
63. **c** From the computer front panel the flight deck annunciations are out of sight. (L3, PDF p. 939)

GPWS testing

PDF pages 941–945 · 7 questions

LEVEL 1 — RECALL

64. An engineer looks at the front of the ground proximity warning computer, so how many status LEDs are fitted there? ●
- a. Five LEDs
 - b. Three LEDs
 - c. Two LEDs

LEVEL 2 — UNDERSTAND AND EXPLAIN

65. When the status LEDs on the front panel of the GPWC turn on
- a. When a fault has been detected
 - b. When the computer is powered
 - c. When a self-test has been started
66. Which colour marks the external fault indication on the computer front panel ●
- a. Green
 - b. Red
 - c. Yellow
67. The green LED on the computer front panel is on, so what does that say? ●
- a. The computer is serviceable
 - b. An external fault is present
 - c. The computer has failed
68. Which colour marks a failure of the computer itself ●
- a. Red
 - b. Yellow
 - c. Green

LEVEL 3 — APPLY AND REASON

69. Level 2 reports No Faults but the engineer must still see what went wrong on earlier flights, so which level is needed?
- a. Level 6
 - b. Level 3
 - c. Level 4
70. The yellow LED is lit on the front panel while the green one stays on, so where should the engineer look first? ●
- a. At the equipment feeding the computer
 - b. At the power supply to the computer
 - c. Inside the computer itself

Answers — GPWS testing

64. **b** Three status LEDs are fitted on the front panel. (L1, PDF p. 945)
65. **b** The LEDs come on whenever the computer has power. (L2, PDF p. 945)
66. **c** The external fault LED is yellow. (L2, PDF p. 945)

- 67. a Green is the computer OK indication. (L2, PDF p. 945)
- 68. a The computer fail LED is red. (L2, PDF p. 945)
- 69. c Current faults are level 2; the history of earlier legs is level 4. (L3, PDF p. 941)
- 70. a Yellow marks an external fault while green still says the computer is good, so the trouble is outside it. (L3, PDF p. 945)

Vibration measurement and indication

PDF pages 947–961 · 41 questions

LEVEL 1 — RECALL

71. What the engine vibration indication lets you identify ●
 - a. A fuel control fault
 - b. An oil filter blockage
 - c. A rotor imbalance
72. An engine vibration reading is seen in the cockpit, so how is its level expressed? ●
 - a. Usually in units
 - b. Usually in millimetres per second
 - c. Usually as a percentage of maximum
73. Which kind of transducer is used as an engine vibration sensor ●
 - a. A strain gauge bridge
 - b. A tachometer generator
 - c. An accelerometer
74. A technician counts the vibration sensors fitted to one engine, so how many does he normally find? ●
 - a. Two
 - b. Four
 - c. One
75. Which two types of accelerometer are used as engine vibration sensors ●
 - a. The electromagnetic type and the piezoelectric-crystal type
 - b. The piezoelectric-crystal type and the strain gauge type
 - c. The electromagnetic type and the variable reluctance type
76. What makes a piezoelectric crystal generate a voltage ●
 - a. A current passed through the crystal
 - b. A force applied to the crystal
 - c. A magnet moved across the crystal
77. A test switch is fitted to an analogue vibration display, so how much of it can be checked? ●
 - a. The sensor wiring only
 - b. The filter setting only
 - c. The entire indicator
78. A glass cockpit display shows the engine vibration, so how many vibration indications are given per engine? ●
 - a. Two
 - b. One
 - c. Four
79. An EICAS display shows the engine vibration, so how many values are given for each engine? ●
 - a. Four
 - b. Two
 - c. One

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 80.** Why a rotor imbalance matters to the engineer ●
- It can be the first sign of engine damage
 - It shows the engine is running overspeed
 - It always means a bearing has already failed
- 81.** What the engine vibration sensors deliver to the computer ●
- Electrical signals
 - Pneumatic pressure signals
 - Hydraulic pressure signals
- 82.** What the computer does with the data it receives from the sensors ●
- It passes it unchanged to the flight data recorder
 - It turns it straight into a cockpit warning indication
 - It filters it for indication and trend monitoring
- 83.** Which other parts carry vibration sensors on many helicopter types? ●
- Flight control actuators
 - Rotors and gearboxes
 - Fuel pumps and filters
- 84.** Which quantity of the rotor system is sensed by an engine vibration pick-up ●
- The radial acceleration of the rotor system
 - The axial displacement of the rotor system
 - The rotational speed of the rotor system
- 85.** One of the two vibration sensors of an engine sits in the compressor area, so where is the other one? ●
- In the turbine area
 - In the accessory gearbox area
 - In the combustion chamber area
- 86.** How is the permanent magnet carried inside an electromagnetic vibration sensor ●
- It is held on a base plate by an inertia mass, with a coil around it
 - It is held in the centre by a fixed coil, with two springs around it
 - It is held in the centre by two springs, with a fixed coil around it
- 87.** The housing of an electromagnetic vibration sensor shakes with the engine, so what does the magnet inside do?
- It moves up and down with the housing
 - It stays almost still because of its inertia
 - It turns with the rotor system
- 88.** What produces the AC output voltage of an electromagnetic vibration sensor ●
- The difference in motion between the coil and the magnetic field
 - The force of the two springs acting on the permanent magnet
 - The pressure of an inertia mass on the base plate
- 89.** What is the job of the attuned filter inside an analogue vibration indicator
- It screens out the undesired frequencies
 - It amplifies the undesired frequencies
 - It rectifies the critical frequencies

90. What does the marking REL AM on a vibration indicator tell the technician ●
- That the scale carries no units and only follows a trend
 - That the scale shows the speed difference of the rotors
 - That the scale is graduated in mils of displacement
91. Which two rotor systems are covered by the two vibration indications of one engine ●
- The N2 rotor system and the accessory drive system
 - The N1 rotor system and the N2 rotor system
 - The N1 rotor system and the fan rotor system
92. From how many sensor signals does the monitoring unit make the two vibration indications ●
- From one single sensor signal on its own
 - From one sensor signal for each indication
 - From the two sensor signals added together
93. Which of the two engine vibration sensors is normally used for this display indication? ●
- Whichever reads higher
 - The aft one
 - The forward one
94. The forward vibration sensor of an engine fails, so what does the monitoring unit do? ●
- It freezes the last indicated value
 - It removes the N2 indication only
 - It switches over to the aft one
95. In which two formats is the vibration value put on an EICAS display ●
- In an analog and a digital format
 - In a digital and a bar graph format
 - In an analog and a colour coded format
96. A technician studies the vibration display, so what is put on it beside the value itself?
- The sensor and filter functions in use
 - The stored limit values for the engine
 - The rotor speeds used for the comparison

LEVEL 3 — APPLY AND REASON

97. The cockpit vibration reading creeps up over several flights with no other symptom, so what does the trend suggest? ●
- The indicator calibration has drifted
 - The sensor wiring is open circuit
 - A rotor imbalance is developing
98. The vibration sensor near the N1 rotor shaft of an engine has failed, so which part of the engine has lost its pick-up? ●
- The accessory gearbox area
 - The turbine area
 - The compressor area

- 99.** A vibration signal reaches the monitoring unit with a high voltage but a low frequency, so what has it been told?
- A small acceleration at a fast vibration
 - A large acceleration at a slow vibration
 - A large rotor speed at a slow acceleration
- 100.** Why can an electromagnetic vibration sensor work without any outside supply ●
- Because relative motion induces its voltage, as in a generator
 - Because its coil is fed from the engine tachometer circuit
 - Because its springs store the energy of each vibration cycle
- 101.** A technician wants the indication of a low frequency vibration on an older system, so which switch does he use?
- The one that picks the engine
 - The one that picks the sensor
 - The one that picks the filter
- 102.** An indicator carries a REL AM scale and two readings are taken a week apart, so what can be done with them? ●
- They can be read as absolute amplitudes
 - They can be read as rotor speed values
 - They can be compared to follow the trend
- 103.** An older vibration system must show the oscillations of the add-on components, so which filter position is used?
- The low frequency one
 - The high frequency one
 - The broad band one
- 104.** An analogue vibration indicator is driven by a moving coil, so what must be done to the alternating sensor signal before it can drive that movement?
- It must be digitised
 - It must be rectified
 - It must be attenuated
- 105.** Why does an analogue vibration indicator need a filter selection at all
- Because motor and add-on vibrations fall in different frequency ranges
 - Because the scale carries no units of measurement, only a trend
 - Because the sensor output is an alternating current, not a direct one
- 106.** The pointer of a vibration indicator sits below the zero mark, so what has happened?
- The vibration has fallen to nothing
 - A system error has occurred
 - The filter has been switched off
- 107.** Why does the vibration monitoring system need the tachometer signals as well as the accelerometer signals
- Because the indicator scale must be set to the right range
 - Because the tracking filter must follow the rotor speed
 - Because the charge amplifiers must be supplied with power

- 108.** How can one sensor signal be turned into separate N1 and N2 vibration indications ●
- By comparing its frequency with the two rotor speed signals
 - By switching the filter between the forward and the aft pick-up
 - By comparing its amplitude with the two stored limit values
- 109.** A rotor speed signal is not available to the monitoring unit, so which indication mode is used?
- BB
 - LPT
 - FAN
- 110.** Which actions may a pilot take after a vibration indication begins to pulse
- Checking the oil pressure, closing the bleeds or re-lighting the engine
 - Selecting the aft pick-up, resetting the filter or changing the mode
 - Scanning other engine indications, reducing power or shutting down
- 111.** Why are the sensor and filter functions displayed beside a single vibration value
- Because each rotor system is given its own separate indication
 - Because the unit picks the highest signal and its source must be known
 - Because the crew choose the sensor and the filter with two switches

Answers — Vibration measurement and indication

- 71. c** The vibration indication reveals a rotor imbalance. (L1, PDF p. 947)
- 72. a** The cockpit indication is normally given in units. (L1, PDF p. 947)
- 73. c** The vibration sensors on an engine are accelerometers. (L1, PDF p. 949)
- 74. a** Two sensors per engine is the usual fit. (L1, PDF p. 949)
- 75. a** Only these two accelerometer types are used on engines. (L1, PDF p. 951)
- 76. b** Applying force to the crystal is what produces the voltage. (L1, PDF p. 951)
- 77. c** The test switch checks the whole indicator. (L1, PDF p. 955)
- 78. a** Two indications are given for each engine. (L1, PDF p. 959)
- 79. c** Only one value per engine is shown on this display. (L1, PDF p. 961)
- 80. a** An imbalance can be the earliest sign that the engine is being damaged. (L2, PDF p. 947)
- 81. a** The sensors send electrical signals to the computer. (L2, PDF p. 947)
- 82. c** The computer collects and filters the data for both indication and trend monitoring. (L2, PDF p. 947)
- 83. b** On many helicopters the rotors and gearboxes also carry vibration sensors. (L2, PDF p. 947)
- 84. a** The accelerometer senses acceleration in the radial direction. (L2, PDF p. 949)
- 85. a** The second of the two sensors is in the turbine area. (L2, PDF p. 949)
- 86. c** Two springs hold the magnet centrally and the coil is the fixed part around it. (L2, PDF p. 951)
- 87. b** Inertia keeps the magnet nearly stationary while the housing and coil move. (L2, PDF p. 951)
- 88. a** Relative motion between coil and field induces the voltage, as in a generator. (L2, PDF p. 951)
- 89. a** The attuned filter removes the frequencies that are not wanted. (L2, PDF p. 955)
- 90. a** REL AM means an unitless scale meant only for trend following. (L2, PDF p. 955)
- 91. b** One indication is for N1 and the other for N2. (L2, PDF p. 959)
- 92. a** Both indications come from a single sensor signal. (L2, PDF p. 959)
- 93. c** The forward sensor is the one normally used. (L2, PDF p. 959)
- 94. c** On failure of the forward sensor the unit changes over to the aft sensor. (L2, PDF p. 959)
- 95. a** The display gives the value both in analog and in digital form. (L2, PDF p. 961)
- 96. a** The sensor and filter functions being used are shown with the value. (L2, PDF p. 961)

- 97. c** A rising trend in the vibration level is read as a developing rotor imbalance, the first sign of damage. (L3, PDF p. 947)
- 98. c** The sensor near the N1 rotor shaft is the one in the compressor area, so that area is the one left unmonitored. (L3, PDF p. 949)
- 99. b** Voltage carries the acceleration level and frequency carries the vibration frequency, so high voltage with low frequency is a big acceleration vibrating slowly. (L3, PDF p. 951)
- 100. a** The sensor generates its own voltage by induction, the way a generator does. (L3, PDF p. 951)
- 101. c** Frequency band is set by the filter switch, not by the sensor switch. (L3, PDF p. 953)
- 102. c** With no units on the scale the readings are only useful compared with each other as a trend. (L3, PDF p. 955)
- 103. a** Add-on components give the low oscillations, so the low frequency filter is selected. (L3, PDF p. 955)
- 104. b** A moving coil instrument needs a unidirectional current, so the alternating signal is rectified before display. (L3, PDF p. 955)
- 105. a** High oscillations come from the motor and low ones from the add-on components, so the filter has to be set to the range being looked at. (L3, PDF p. 955)
- 106. b** A pointer below zero on the OFF / Error mark means a system error, not a zero vibration. (L3, PDF p. 957)
- 107. b** The filter has to track the rotor speed, and it is the tachometer signals that tell it that speed. (L3, PDF p. 957)
- 108. a** The frequency of the vibration signal is compared with the two rotor speed signals to sort it into N1 and N2. (L3, PDF p. 959)
- 109. a** Without a rotor speed signal the filters cannot track, so the unfiltered broad band mode is used. (L3, PDF p. 961)
- 110. c** The pilot has to look for the reason, by scanning other engine indications, reducing power or shutting the engine down. (L3, PDF p. 961)
- 111. b** With only one value shown and the unit free to pick whichever signal is highest, the crew must be told which sensor and filter produced it. (L3, PDF p. 961)

Rotor and transmission monitoring

PDF pages 963–967 · 23 questions

LEVEL 1 — RECALL

- 112.** Which three components make up the main rotor rpm monitoring system ●
- a. A magnetic pick up, an rpm indicator and an rpm warning unit
 - b. A magnetic pick up, an rpm indicator and a triple tachometer
 - c. A magnetic pick up, an rpm warning unit and a master caution light
- 113.** A technician must find the rotor rpm magnetic pick up, so where is it installed? ●
- a. Within the tail rotor gearbox
 - b. Within the main gear box
 - c. Within the rotor mast
- 114.** Which three components make up the mast moment monitoring system ●
- a. A strain gauge bridge, a magnetic pick up and indicators
 - b. A strain gauge bridge, a sensor transmitting unit and indicators
 - c. A sensor transmitting unit, an rpm warning unit and indicators
- 115.** A strain gauge bridge is fitted for the bending moment monitoring, so where does it sit? ●
- a. In the rotor head
 - b. In the rotor mast
 - c. In the main transmission

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 116.** Which kind of sensor measures the main rotor speed ●
- a. A magnetic type one
 - b. A piezoelectric type one
 - c. A capacitive type one
- 117.** What is the voltage peak from the rotor speed pickup proportional to ●
- a. The temperature of the main gear box
 - b. The movement of the interrupter
 - c. The frequency of the interrupter
- 118.** A pilot reads the rotor rpm on the triple tachometer, so which part of the dial shows it? ●
- a. The digital window at the foot of the dial
 - b. The large pointer N1 on the outer scale
 - c. The small pointer R on the inner scale
- 119.** The rotor rpm is shown permanently in two places in the cockpit, so which two are they? ●
- a. The triple tachometer and the caution and advisory display
 - b. The triple tachometer and the central engine display unit
 - c. The rpm warning unit and the central engine display unit
- 120.** Through which unit do the speed pickup signals pass on their way to the indicators
- a. The RPM warning unit
 - b. The central engine display unit
 - c. The sensor transmitting unit

- 121.** The rotor rpm falls below the operating range, so what is given to the crew? ●
- A visual and an audio warning
 - A visual warning only
 - A removal of the rpm indication
- 122.** Where are the sampled transmission data shown constantly
- On the triple tachometer or on the rpm warning unit
 - On the analog instruments or on the mast moment indicator
 - On the analog instruments or on the Caution and Advisory Display
- 123.** Which two transmission parameters have their own instruments showing them constantly ●
- The oil temperature and the chip detection
 - The oil pressure and the vibration level
 - The oil temperature and the oil pressure
- 124.** A sampled transmission value goes outside its operational range, so what happens? ●
- A warning appears on the respective warning unit
 - The parameter is taken off the display
 - The indication is frozen at the limit value
- 125.** What does the mast moment monitoring system measure ●
- The bending moment between the rotor mast and the main transmission
 - The vibration between the rotor mast and the rotor head
 - The twisting moment between the engine and the main transmission
- 126.** Through which unit does the measured mast moment reach the indicators ●
- The sensor transmitting unit
 - The caution and advisory display
 - The rpm warning unit
- 127.** The maximum mast moment is exceeded in flight, so what comes with the warning light?
- An entry in the fault memory
 - An aural indication
 - An automatic power reduction
- 128.** How is an exceeded mast moment shown on an electronic mast moment indicator ●
- By a flashing scale
 - By a warning light
 - By a red marking
- 129.** A helicopter is not fitted with an EFIS, so what must be done to the mast moment indicator before each engine start? ●
- It must be calibrated
 - It must be zeroed by the pilot
 - It must be tested against its warning light
- 130.** A helicopter does carry an EFIS, so how is that mast moment indicator check carried out? ●
- It is done by the pilot before each engine start
 - It is done automatically by the system
 - It is done by maintenance at the daily check

LEVEL 3 — APPLY AND REASON

- 131.** A tail rotor gearbox develops a malfunction, so how will the crew learn of it? ●
- By its own analog instrument
 - By a pointer leaving the green arc
 - By a caution light only
- 132.** Why is a main transmission fault easier to judge than a tail rotor gearbox fault? ●
- Both carry constant indications but only the gearbox has a light
 - The transmission carries only a light, the gearbox constant indications
 - The transmission carries constant indications, the gearbox only a light
- 133.** Why can a bending load in the rotor mast be turned into a cockpit indication at all? ●
- Because a magnetic pick up in the mast senses the bending
 - Because a capacitive probe in the mast senses the bending
 - Because a strain gauge bridge in the mast senses the bending
- 134.** A helicopter without EFIS was started without the bending load indicator being calibrated, so what is now in doubt? ●
- The aural warning of the rotor rpm
 - The accuracy of the mast moment indication
 - The supply to the strain gauge bridge

Answers — Rotor and transmission monitoring

- 112. a** These three are the components of the system. (L1, PDF p. 963)
- 113. b** It sits inside the main gear box. (L1, PDF p. 963)
- 114. b** These three are the components listed for the system. (L1, PDF p. 967)
- 115. b** The bridge is installed in the rotor mast. (L1, PDF p. 967)
- 116. a** The rotor speed pickup is a magnetic sensor. (L2, PDF p. 963)
- 117. c** The peak is in proportion to the interrupter frequency. (L2, PDF p. 963)
- 118. c** Rotor rpm is the small R pointer on the inner scale. (L2, PDF p. 963)
- 119. b** Both the triple tachometer and the central engine display unit carry the rotor rpm. (L2, PDF p. 963)
- 120. a** The signals go through the rotor RPM warning unit. (L2, PDF p. 963)
- 121. a** Going outside the range either way sets off both visual and audio warning. (L2, PDF p. 963)
- 122. c** Either the analog indicators or the CAD carry them constantly. (L2, PDF p. 965)
- 123. c** Transmission oil temperature and pressure are the two indicated constantly. (L2, PDF p. 965)
- 124. a** Out-of-range data produce a warning on the respective warning unit. (L2, PDF p. 965)
- 125. a** It is the bending moment between mast and main transmission. (L2, PDF p. 967)
- 126. a** The signal passes through the sensor transmitting unit. (L2, PDF p. 967)
- 127. b** An aural indication accompanies the visual one. (L2, PDF p. 967)
- 128. c** On the electronic indicator it is a red marking; the warning light belongs to the analog one. (L2, PDF p. 967)
- 129. a** Without an EFIS the indicator has to be calibrated before each engine start. (L2, PDF p. 967)
- 130. b** With an EFIS fitted the check is done automatically. (L2, PDF p. 967)
- 131. c** Gearboxes other than the main transmission are only covered by caution lights. (L3, PDF p. 965)
- 132. c** Main transmission values are indicated all the time, while the other gearboxes give nothing but a caution light. (L3, PDF p. 965)
- 133. c** The strain gauge bridge in the mast is what converts the bending into a signal. (L3, PDF p. 967)
- 134. b** Without the pre-start calibration the indicated mast moment can no longer be trusted. (L3, PDF p. 967)

Fuel quantity indication

PDF pages 969–977 · 19 questions

LEVEL 1 — RECALL

135. Which two methods are basically used to measure the fuel quantity ●
- A capacitor system and a float switch system
 - An ultrasonic system and a flow meter system
 - A capacitor system and an ultrasonic system
136. What decides the number of fuel quantity probes in a tank ●
- The size and shape of the aircraft
 - The size and shape of that tank
 - The number of pumps in that tank
137. Which electrical component is a fuel quantity probe ●
- An inductor
 - A resistor
 - A capacitor
138. What is the material dependent factor that affects a probe's capacitance called ●
- The capacitive reactance
 - The dielectric constant
 - The specific gravity

LEVEL 2 — UNDERSTAND AND EXPLAIN

139. A large aeroplane carries an extra fuel quantity indication outside the cockpit, so who needs it?
- The loadmaster in the hold
 - The fuelling personnel
 - The flight crew on the walk round
140. On which aeroplanes is the ultrasonic fuel measuring principle used ●
- On all of them
 - On smaller ones only
 - On larger ones only
141. How are the fuel quantity probes installed in a tank ●
- Vertically, from the top down to about half depth
 - Vertically, from the bottom right up to the top
 - Horizontally, lying along the floor of the tank
142. A technician lays out the probes taken from one tank, so what does he notice about them? ●
- They are all of the same length
 - They are of different length
 - They are of different diameter
143. The amount of fuel in a tank changes, so what changes at the probe? ●
- Its inductance
 - Its capacity
 - Its resistance

- 144.** How are the two tubes of a fuel quantity probe arranged
- Coaxially and electrically bonded to each other
 - Side by side and electrically bonded to each other
 - Coaxially and electrically isolated from each other
- 145.** What is the capacitance of a fuel quantity probe ●
- Its ability to generate a voltage
 - Its capability to store electrical energy
 - Its opposition to the flow of current
- 146.** On which two features of the plates does the capacitance of a probe depend ●
- Their area and the thickness of them
 - Their length and the thickness of them
 - Their area and the distance between them

LEVEL 3 — APPLY AND REASON

- 147.** A small aeroplane is being fitted with a fuel quantity system, so which of the two measuring methods is open to it? ●
- Either of the two methods
 - The one using capacitors
 - The one using the ultrasonic principle
- 148.** A tank is deep and awkwardly shaped, so what does that mean for the probes put in it? ●
- One of them, laid flat at several levels
 - Several of them, cut to different lengths
 - Several of them, all cut to one length
- 149.** The fuel level rises past the middle of a probe, so what has happened between its two tubes? ●
- The gap between the tubes has become smaller
 - All of the air has been replaced by fuel
 - Part of the air has been replaced by fuel
- 150.** Two probes in the same tank are both completely covered by fuel yet give different values, so what explains it? ●
- Their plates differ in area because they differ in length
 - Their plates differ in spacing because they differ in fuel cover
 - Their plates differ in area because they differ in fuel cover
- 151.** The material between a probe's tubes changes from air to fuel, so what does the measuring system see? ●
- A lower capacitance
 - A higher capacitance
 - An unchanged capacitance
- 152.** Why does a capacitive probe give a different reading as the tank fills ●
- Because the spacing between its tubes changes
 - Because the material between its tubes changes
 - Because the area of its tubes changes

- 153.** Why does a half covered probe sit between the empty and the full capacitance value ●
- Because the gap between its tubes has partly been closed up
 - Because only part of its plate area is still in the circuit
 - Because only part of the air between its tubes has been replaced

Answers — Fuel quantity indication

- 135. c** Capacitors and the ultrasonic principle are the two methods given. (L1, PDF p. 969)
- 136. b** Tank size and shape set how many probes are needed. (L1, PDF p. 975)
- 137. c** Each probe is a capacitor. (L1, PDF p. 975)
- 138. b** It is named the dielectric constant. (L1, PDF p. 977)
- 139. b** The indication at the fuelling station is there for the fuelling personnel. (L2, PDF p. 969)
- 140. c** The ultrasonic method is restricted to larger aeroplanes. (L2, PDF p. 969)
- 141. b** They stand vertically and run the full depth of the tank. (L2, PDF p. 975)
- 142. b** Because the tank depth varies the probes come in different lengths. (L2, PDF p. 975)
- 143. b** The probe's capacity follows the amount of fuel. (L2, PDF p. 975)
- 144. c** They sit one inside the other and are isolated, which is what makes them a capacitor. (L2, PDF p. 977)
- 145. b** Capacitance is the ability to store electrical energy. (L2, PDF p. 977)
- 146. c** Plate area A and plate separation d are the two given. (L2, PDF p. 977)
- 147. b** The ultrasonic method is for larger aeroplanes only, which leaves the capacitor method. (L3, PDF p. 969)
- 148. b** Size and shape decide how many probes are needed, and because they must reach from bottom to top they end up different lengths. (L3, PDF p. 975)
- 149. c** The gap is filled to the same level as the tank, so a half covered probe has fuel in part of it and air in the rest. (L3, PDF p. 977)
- 150. a** Probes in one tank are of different lengths, so their plate areas differ even when both are fully covered. (L3, PDF p. 977)
- 151. b** Fuel in place of air raises the capacitance, because its dielectric constant is the higher one. (L3, PDF p. 977)
- 152. b** Filling the tank swaps air for fuel between the tubes, and that material change alters the capacitance. (L3, PDF p. 977)
- 153. c** Fuel raises the capacitance only over the part of the probe it covers, so a half covered probe falls between the two extremes. (L3, PDF p. 977)

Capacitance probes and compensation

PDF pages 979–989 · 51 questions

LEVEL 1 — RECALL

- 154.** Which units are used to show the fuel quantity on a large commercial aircraft ●
- Units of weight
 - Units of volume
 - Units of pressure
- 155.** Which property of air and fuel the capacitance method of quantity measurement makes use of ●
- Their differing electrical resistances
 - Their differing dielectric constants
 - Their differing boiling points
- 156.** A bridge transformer has a centre tap and two secondary windings, so what phase relationship do the windings have to the tap? ●
- Opposite
 - Ninety degrees apart
 - Identical
- 157.** How the immersion depth is measured in the fuel quantity computer system ●
- By a float and an LVDT at the tank units and the compensator
 - By travel time of a pulse at the tank units and the compensator
 - By capacity changes at the tank units and the compensator
- 158.** Which two quantities the cadensicon sensor measures separately for the computer ●
- The density and the dielectric constant
 - The volume and the dielectric constant
 - The temperature and the specific gravity
- 159.** Which four items make up the fuel quantity indicating system that compensates fuel scatter ●
- Tank units, a computer, a densitometer and a water detector
 - Tank units, a computer, a cadensicon and a compensator
 - Tank units, a totaliser, a repeater and a compensator
- 160.** A technician looks for the compensator inside a fuel tank, so where in the tank is it installed? ●
- At about mid height
 - Near the lowest point
 - Near the highest point

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 161.** Why the amount of fuel carried is indicated in units of weight ●
- The energy available from the fuel depends on weight
 - The energy available from the fuel depends on volume
 - The tank capacity is certified in terms of weight
- 162.** What makes a direct weight indication of the fuel necessary on a large commercial aircraft ●
- Fuel can be half the take-off weight and the approved weights need it
 - Fuel is always loaded by volume and the tank gauges are marked in litres only
 - Fuel density is constant so the weight can be read straight from the volume

- 163.** The same number of litres of fuel is loaded on a very cold day and on a very hot day, so what differs between the two loads?
- The weight on board
 - The capacity of the empty tank
 - The dielectric constant of the fuel
- 164.** The fuel level in a tank alters, so what does the multi-tubular surface condenser in it do? ●
- It changes its resistance
 - It changes its excitation frequency
 - It changes its capacity
- 165.** What the total capacity of the tank condenser is made up of ●
- The empty tank capacity minus the change from fuel displacing air
 - The empty tank capacity plus the change from fuel displacing air
 - The change from fuel displacing air on its own
- 166.** What happens to the capacitance of a tank sensor as the tank fills with fuel ●
- It decreases
 - It increases
 - It stays constant
- 167.** A tank sensor has changed its capacitance as the tank filled, so where is that change used? ●
- In a bridge circuit that gives an amplified signal
 - In a resolver circuit that gives a shaft angle output
 - In a chopper circuit that gives a pulse train output
- 168.** What the amplified signal from the capacitance bridge acts on ●
- The pointer or the digital display
 - The shutoff valve in the refuelling gallery line
 - The excitation winding of the tank unit sensor
- 169.** What condition exists when the tank unit capacitance and the reference condenser capacitance are equal ●
- No current runs through the instrument
 - Maximum current runs through the instrument
 - A current runs that drives the pointer to full
- 170.** What is true of the two points across which the instrument is connected when the bridge is balanced ●
- They are at the supply potential
 - They are at opposite electrical potential
 - They are at the same electrical potential
- 171.** What the position of the potentiometer arm represents once the bridge has been rebalanced ●
- The supply voltage applied to the bridge
 - The ratio of the two capacitance values
 - The sum of the two capacitance readings
- 172.** How the capacitive bridge of a fuel quantity indicator is rebalanced in service ●
- Automatically, by amplifying the difference signal
 - Automatically, by switching in reference condensers
 - Manually, by the crew turning a trimming knob

- 173.** The amplifier of a capacitive fuel gauge puts out a voltage, so what does that voltage drive? ●
- A motor that turns the potentiometer
 - A solenoid that trips the warning flag
 - A heater that trims the reference condenser
- 174.** Fuel is being burned off in flight and the gauge is following it down, so what brings the rebalancing motor to rest at each new reading? ●
- The amplifier output reaching its limit
 - The difference signal falling to zero
 - The potentiometer reaching its end stop
- 175.** How the bridge turns a change of tank capacitance into something the indicator can show ●
- The secondary winding phase is taken as the indication
 - The instrument current is taken straight as the indication
 - The rebalanced potentiometer position is indicated
- 176.** Why the relationship between dielectric constant and density has to be corrected ●
- It varies with the fuel consistency of each country
 - It varies with the age of the tank units in the tank
 - It varies with the altitude the aircraft is flown at
- 177.** How the dielectric constant is obtained inside a cadensicon sensor ●
- From the depth of immersion of a float
 - From the capacity change of a condenser
 - From the travel time of an ultrasonic pulse
- 178.** How the density is obtained inside a cadensicon sensor ●
- By a condenser whose capacity changes with the fuel
 - By a thermistor bridge sensing the fuel temperature
 - By a cardanically suspended float of fixed weight
- 179.** The cadensicon float has sunk to a certain depth in the fuel, so how is that depth turned into a signal for the computer? ●
- By a synchro resolver
 - By a strain gauge
 - By an LVDT
- 180.** How density is found in systems that damp a mass instead of floating one ●
- From the force needed to hold the damped mass still
 - From the ratio of oscillations added to those removed
 - From the time the damped mass takes to come to rest
- 181.** What the fuel quantity indicating computer works out from the probes it monitors
- The mass of fuel in the related tank
 - The volume of fuel in the related tank
 - The density of fuel in the related tank
- 182.** When the cadensicon sensor takes its measurements of K and D ●
- Only when the tank is nearly empty
 - As fuel enters the tank
 - Only once the engines have been started

- 183.** In what form the cadensicon delivers its two outputs to the computer ●
- Both as electrical signals
 - One electrical and one pneumatic
 - Both as mechanical movements
- 184.** A cadensicon sensor measures two properties of the fuel, so which one of them is sensed mechanically? ●
- The temperature
 - The dielectric constant
 - The density
- 185.** What the fuel scatter effect is ●
- The capacitance of a probe drifts as the probe gets older
 - The K to D relationship alters as the fuel temperature changes
 - The K to D relationship is not constant between fuel samples
- 186.** The fuel in a tank warms up over a long flight, so what happens to the relationship between K and D? ●
- It is pulled back onto the average law
 - It is not influenced by temperature
 - It changes in step with the fuel density

LEVEL 3 — APPLY AND REASON

- 187.** Water has collected between the plates of a capacitance probe, so how does the probe capacitance compare with the same probe covered in fuel? ●
- The same
 - Slightly lower
 - Much higher
- 188.** Water lying in a tank gives a far greater capacitance than fuel does, so how would the quantity indication be affected? ●
- It would read above the fuel actually present
 - It would read below the fuel actually present
 - It would stay at the correct reading
- 189.** Why the capacity of a tank condenser is compared with its capacity when the tank is empty ●
- The difference corresponds to the quantity of fuel present
 - The difference corresponds to the dielectric constant of the fuel
 - The difference corresponds to the temperature of the fuel
- 190.** What is gained by leaving the tank unit as the only variable in the bridge
- A lower supply current to the bridge
 - A faster response of the indication
 - Greater precision of the indication
- 191.** The aircraft is being refuelled and the level in a tank is rising, so what is happening at the master indicator?
- Compensator current rises and the indication holds steady
 - Tank unit current falls and the indication moves downscale
 - Tank unit current rises and the indication moves upscale

- 192.** Why the indicator runs downscale when the fuel quantity test switch is operated
- The supply to the rebalancing motor is reversed
 - The compensator current that normally drives it to full is missing
 - The tank unit current that normally drives it to full is missing
- 193.** A coaxial cable to one tank unit has broken so that no tank unit current reaches its master indicator, so what will that indicator do?
- Freeze at the last reading
 - Run upscale to full
 - Run downscale
- 194.** The rebalancing motor of a capacitive gauge has just run to a stop, so what is true of the two bridge currents at that moment?
- They are equal
 - They are both at zero
 - They are in opposite phase
- 195.** The master indicator already shows the figure it has worked out, so how do the repeater and the fuel summation unit come to show the same?
- The computed value is sent to them over the compensator line
 - The tank unit current is sent to them over the coaxial cable
 - The synchronising voltage is sent to them from the totaliser
- 196.** How the built-in test of the computerised system differs from the test started by the flight deck switch
- One runs automatically and codes faults, the other tests the displays
 - One runs automatically and tests the displays, the other codes faults
 - One tests the displays, the other codes faults found on the ground only
- 197.** A computerised fuel quantity indicator is showing an error code instead of a quantity, so which of the computer circuits produced it?
- The one running the automatic system check
 - The one feeding the repeater and the summation unit
 - The one sending the signal out to the tank units
- 198.** A float sinks further in one fuel sample than in another of the same temperature, so what does that tell the computer? ●
- That the two samples differ in volume
 - That the two samples differ in density
 - That the two samples differ in dielectric constant
- 199.** Refuelling has to stop before a tank overflows, so what is built into the fuel quantity computer system for this? ●
- A totaliser that sums all of the master systems
 - A preselection of the wanted load with a shutoff
 - A trimming condenser at the cockpit indicator
- 200.** Why the aircraft attitude matters to the preselection and shutoff function ●
- The density measured by the float depends on aircraft position
 - The dielectric constant of the fuel depends on aircraft position
 - The point at which a tank overflows depends on aircraft position

- 201.** What the computer derives from the cadensicon and compensator data it monitors
- The density of the fuel in the tanks
 - The temperature of the fuel in the tanks
 - The volume of the fuel in the tanks
- 202.** The probes alone can only tell the computer how much space the fuel fills, so what else must it have before a figure in kilograms can be shown?
- The temperature of the fuel
 - The usable capacity of the tank
 - The density of the fuel
- 203.** The outer tanks still hold fuel from the previous flight when fresh fuel is loaded, so how does the computer handle the mixture?
- It ignores the old fuel until that tank has been emptied
 - It calculates a mean value from the old and the new fuel
 - It applies the values of the fresh fuel to both quantities
- 204.** Why fuel taken on at two different airports can upset the quantity indication unless the computer allows for it
- The two loads can have different fuel temperatures
 - The two loads can have different K to D values
 - The two loads can have different tank unit capacitances

Answers — Capacitance probes and compensation

- 154. a** On large commercial aircraft the fuel quantity is shown in units of weight. (L1, PDF p. 979)
- 155. b** The capacitance method uses the differing dielectric constants of air and fuel. (L1, PDF p. 979)
- 156. a** The two secondary windings are in opposite phase to the centre tap. (L1, PDF p. 981)
- 157. c** Immersion depth is measured by capacity changes using the tank units and the compensator. (L1, PDF p. 987)
- 158. a** The cadensicon measures density and dielectric constant separately. (L1, PDF p. 987)
- 159. b** The system has tank units, a fuel quantity indicating computer, a cadensicon sensor and a compensator. (L1, PDF p. 989)
- 160. b** The compensators are installed near to the lowest point of the tanks. (L1, PDF p. 989)
- 161. a** The energy present in the form of fuel is dependent on weight, so weight is indicated. (L2, PDF p. 979)
- 162. a** Fuel can be up to one half of take-off weight, so direct weight indication is needed for the approved take-off and landing weights. (L2, PDF p. 979)
- 163. a** Volume and specific gravity change with temperature, so the same number of litres can differ by several tonnes. (L2, PDF p. 979)
- 164. c** The condenser changes its capacity depending on the level of fuel in the tank. (L2, PDF p. 979)
- 165. b** Total capacity is the empty tank capacity plus the capacity change due to the displacement of air between the plates. (L2, PDF p. 979)
- 166. b** Fuel has a dielectric strength more than twice that of air, so sensor capacitance rises with fuel content. (L2, PDF p. 979)
- 167. a** The change in capacitance is used in a bridge circuit to provide an amplified signal. (L2, PDF p. 979)
- 168. a** The amplified signal rotates the pointer or actuates the digital display in the indicator. (L2, PDF p. 979)
- 169. a** With equal capacitances the two currents compensate each other and nothing flows through the instrument. (L2, PDF p. 981)
- 170. c** At balance the two points have the same electrical potential, so no current flows. (L2, PDF p. 981)
- 171. b** The potentiometer arm position indicates the ratio of the two capacitance values. (L2, PDF p. 981)
- 172. a** Fuel quantity capacitive bridges are balanced automatically by amplifying the difference signal. (L2, PDF p. 981)
- 173. a** The amplifier output voltage drives a motor which turns the potentiometer. (L2, PDF p. 981)

- 174. b** The motor stops when the difference signal is zero. (L2, PDF p. 981)
- 175. c** The bridge is rebalanced by the motor-driven potentiometer, and the motor or potentiometer position is what is indicated. (L2, PDF p. 981)
- 176. a** The ratio of K to D varies in accordance with fuel consistency in individual countries. (L2, PDF p. 987)
- 177. b** K is calculated from the capacity change of a condenser. (L2, PDF p. 987)
- 178. c** D is found mechanically by a cardanically suspended float of predetermined weight. (L2, PDF p. 987)
- 179. c** The depth of immersion is measured by an LVDT and sent to the computer. (L2, PDF p. 987)
- 180. b** In those systems the ratio of oscillations added to oscillations removed determines density. (L2, PDF p. 987)
- 181. b** The computer uses signal conditioning on the probe signals to calculate the volume of fuel in the related tank. (L2, PDF p. 989)
- 182. b** The cadensicon measures K and D separately as fuel enters the tank. (L2, PDF p. 989)
- 183. a** Outputs K and D are both electrical signals. (L2, PDF p. 989)
- 184. c** K is measured electrically only while D is measured mechanically by a float. (L2, PDF p. 989)
- 185. c** Fuel scatter is the phenomenon that the relationship of K to D between different fuel samples is not constant. (L2, PDF p. 989)
- 186. b** A change in temperature will not influence this relationship. (L2, PDF p. 989)
- 187. c** Water is given a far larger dielectric constant than fuel, and the capacitance rises with the dielectric constant of whatever fills the probe. (L3, PDF p. 979)
- 188. a** The capacitance follows the dielectric constant of what fills the probe, and water's is far above that of fuel, so the gauge reads more fuel than is there. (L3, PDF p. 979)
- 189. a** The capacity above the empty tank value is the part caused by fuel, and that difference corresponds to the fuel quantity. (L3, PDF p. 979)
- 190. c** Having only the tank unit variable increases indicator precision. (L3, PDF p. 983)
- 191. c** A rising level raises the tank unit capacitance and current, and the quantity indicator moves upscale. (L3, PDF p. 983)
- 192. c** Operating the test switch removes the tank unit current that normally moves the indicator towards FULL, so it runs downscale. (L3, PDF p. 983)
- 193. c** A missing tank unit current is what the test switch creates, and it drives the indicator downscale. (L3, PDF p. 983)
- 194. a** The motor runs until the two currents are equal, and that is where it stops. (L3, PDF p. 983)
- 195. a** The value shown on the master indicator is transmitted in a further circuit via the compensator line to the repeater and the fuel summation unit. (L3, PDF p. 985)
- 196. a** The built-in test runs automatically and shows any fault as an error code, while the FQ test switch starts a display test of all indicators. (L3, PDF p. 985)
- 197. a** The error code comes from the separate circuit that runs the automatic system check by built-in test. (L3, PDF p. 985)
- 198. b** The float of predetermined weight submerges in accordance with the density, so a deeper float means a different density. (L3, PDF p. 987)
- 199. b** The system integrates a preselection of the desired volume together with a shutoff on reaching it or shortly before overflowing. (L3, PDF p. 987)
- 200. c** Shutoff occurs on reaching the preselected value or shortly before overflowing, which depends on aircraft position. (L3, PDF p. 987)
- 201. a** Cadensicon and compensator data are used to calculate the density of the fuel in the tanks. (L3, PDF p. 989)
- 202. c** The computer needs both the volume and the fuel density before it can calculate the fuel mass. (L3, PDF p. 989)
- 203. b** The computer can calculate a mean value from the remaining fuel and the replenished fuel. (L3, PDF p. 989)
- 204. b** Fuel consistency differs between countries, so the remaining and replenished fuel can have different K to D values that the computer has to average. (L3, PDF p. 989)

Ultrasonic measurement, densitometers and testing

PDF pages 991–995 • 23 questions

LEVEL 1 — RECALL

- 205.** Which kind of signal the tank units of this liquid level metering system transmit ●
- An ultrasonic signal
 - A light pulse
 - A radio frequency signal
- 206.** A tank of an ultrasonic system is opened for inspection, so how many tank units will be found in it? ●
- A single unit
 - Numerous units
 - Exactly two units
- 207.** A technician surveys an ultrasonic fuel tank installation, so how many densitometers will each tank have? ●
- One
 - One for every tank unit
 - Two
- 208.** Where the water detectors are fitted ●
- At the highest point of each fuel tank
 - At the low point in each fuel tank
 - In the refuelling gallery pipework
- 209.** Which kind of quantity measuring systems the GF-2 fuel tester is used on ●
- Float operated gauging systems
 - Capacitance measuring systems
 - Ultrasonic measuring systems
- 210.** Which three functions the tester for capacitance quantity systems provides ●
- Measuring capacities, simulating capacities and measuring contact resistances
 - Measuring resistances, simulating voltages and measuring tank unit currents
 - Measuring capacities, simulating densities and measuring insulation resistances

LEVEL 2 — UNDERSTAND AND EXPLAIN

- 211.** Where the signal sent out by an ultrasonic tank unit is reflected ●
- At the wall of the stillwell tube
 - At the bottom of the tank sump
 - At the surface of the liquid
- 212.** What picks up the echo in an ultrasonic quantity metering system ●
- The compensator near the tank bottom
 - A separate receiver at the tank top
 - The tank unit that sent the signal
- 213.** A computer has received the echo from an ultrasonic tank unit, so what does it use to find the liquid level? ●
- The running time of the signal
 - The strength of the returned echo
 - The frequency shift of the returned echo

- 214.** What the fuel quantity processor unit measures in order to find the fuel height ●
- The frequency of the returned pulse
 - The travel time of the pulse
 - The strength of the returned pulse
- 215.** Through what the sonic pulse travels on its way to the fuel surface ●
- The wall of the tank unit
 - The air space above the fuel
 - The fuel in the stillwell
- 216.** Some tank units also carry calibration targets, so what does the processor unit get from the pulses that come back off them?
- The velocity of sound in the fuel
 - The density of the fuel in the tank
 - The depth of water in the tank sump
- 217.** Which kind of transducer a densitometer uses ●
- A vibrating cylinder
 - A capacitive condenser
 - A cardanically suspended float
- 218.** A densitometer output frequency is being read off, so what is that frequency proportional to? ●
- The fuel temperature
 - The fuel height
 - The fuel density
- 219.** How a water detector shows that water is lying in the bottom of a tank ●
- Its float sinks in the water and drives a transformer
 - Its pulse is absorbed so that no echo returns at all
 - Its pulse reflects off the boundary with the fuel
- 220.** A technician is told the water detectors work in the same way as the tank units, so what kind of devices are they? ●
- Ultrasonic devices
 - Resistive devices
 - Capacitive devices
- 221.** Which part of the test equipment must be kept clear of the airframe during a capacitance check
- The connector of the tank unit
 - The casing of the tester
 - The shielding of the coaxial cable

LEVEL 3 — APPLY AND REASON

- 222.** Which property of the fuel the capacitance of a tank unit responds to, unlike an ultrasonic tank unit ●
- Its dielectric constant
 - Its temperature in flight
 - Its density in the tank

- 223.** An ultrasonic system has to display a fuel weight, so which two measurements must the processor unit bring together? ●
- The fuel density and the velocity of sound
 - The fuel height and the fuel temperature
 - The fuel height and the fuel density
- 224.** How the ultrasonic system obtains the density it needs, compared with a capacitive densicon system ●
- From a float and a condenser instead of a vibrating cylinder
 - From the pulse travel time instead of a vibrating cylinder
 - From a vibrating cylinder instead of a float and a condenser
- 225.** A densitometer in one tank of an ultrasonic system has failed, so which calculation is lost first? ●
- The water level in that tank
 - The fuel height in that tank
 - The fuel weight for that tank
- 226.** A capacitance quantity system has to be checked on the ground with the tanks drained, so what does the tester let the technician do instead of wetting the probes? ●
- Measure the fuel density
 - Simulate the capacities
 - Measure the contact resistances
- 227.** A tank unit reads correctly on the tester but the cockpit indication is still wrong, so which further tester function would be used on the wiring? ●
- Measuring the tank unit currents
 - Simulating the probe capacities
 - Measuring contact resistances

Answers — Ultrasonic measurement, densitometers and testing

- 205. a** The tank units transmit an ultrasonic signal. (L1, PDF p. 991)
- 206. b** Each tank contains numerous tank units. (L1, PDF p. 991)
- 207. a** Each fuel tank has one densitometer. (L1, PDF p. 993)
- 208. b** Water detectors are at the low point in each fuel tank. (L1, PDF p. 993)
- 209. b** The tester is used for capacitance measuring systems for fuel and oil. (L1, PDF p. 995)
- 210. a** The tester measures capacities, simulates capacities and measures contact resistances. (L1, PDF p. 995)
- 211. c** The signal is reflected at the surface of the liquid. (L2, PDF p. 991)
- 212. c** The same tank unit that transmits also receives the echo. (L2, PDF p. 991)
- 213. a** The computer calculates the liquid level from the running time of the signal. (L2, PDF p. 991)
- 214. b** The processor unit measures the travel time of the pulse and uses it to find the fuel height. (L2, PDF p. 993)
- 215. c** The pulse goes through the fuel in the stillwell to the fuel surface and back to the receiver. (L2, PDF p. 993)
- 216. a** Pulses reflected from the targets are used to measure the velocity of sound in the fuel. (L2, PDF p. 993)
- 217. a** Densitometers use a vibrating cylinder transducer. (L2, PDF p. 993)
- 218. c** The frequency of the vibrations is proportional to the fuel density. (L2, PDF p. 993)
- 219. c** If water is present the pulse reflects off the interface between the water and the fuel. (L2, PDF p. 993)
- 220. a** The water detectors are ultrasonic devices like the tank units. (L2, PDF p. 993)
- 221. b** The tester casing must never come into contact with the airframe. (L2, PDF p. 995)
- 222. a** Capacitive measurement works from the differing dielectric constants, while the ultrasonic unit works from a reflected signal. (L3, PDF p. 991)

- 223. c** Height comes from the tank units and density from the densitometer, and the FQPU calculates the fuel weight from them. (L3, PDF p. 993)
- 224. c** The ultrasonic system gets density from a vibrating cylinder densitometer, whereas the cadensicon gets it from a float with K from a condenser. (L3, PDF p. 993)
- 225. c** Height still comes from the tank unit, but without the density from the densitometer the processor unit cannot work out that tank's fuel weight. (L3, PDF p. 993)
- 226. b** The tester can simulate capacities as well as measure them. (L3, PDF p. 995)
- 227. c** Besides measuring and simulating capacities the tester measures contact resistances, which is what a wiring check needs. (L3, PDF p. 995)